HARP-NET CKM: Hybrid Deep Learning for Dense UAV Channel Knowledge Maps

Guillem Moreno Garcia, Evgenii Vinogradov, and Sergi Abadal

***Abstract*—Channel knowledge maps (CKMs) promise fast environment aware radio planning, but dense UAV CKM generation remains difficult when channel attenuation, delay spread, and angular spread must all generalize to unseen cities. This paper presents HARP-NET CKM (Height-Aware Residual Prior Network for Channel Knowledge Map prediction), a hybrid surrogate that freezes physical and statistical priors calibrated on the training split and learns bounded residual maps with a probabilistic residual head. The model predicts 513×513 dense channel attenuation, DS, and AS maps from topology, geometric masks, UAV transmitter height, and frozen priors through a probabilistic residual stage followed by deterministic RMSE oriented fine tuning. On held out cities it reaches 1.6519 dB channel attenuation RMSE, 26.5570 ns delay spread RMSE, and 11.3854° angular spread RMSE, reducing RMSE relative to the frozen priors for all three targets. The result supports a hybrid design principle: freeze stable radial, visibility, height, and spread structure in auditable priors, and reserve neural capacity for the residual distribution those priors cannot explain.**

***Index Terms*—channel knowledge maps, UAV communications, radio map prediction, channel attenuation, delay spread, angular spread, Gaussian mixture models, physics informed learning.**

I. INTRODUCTION AND RELATED WORK

Channel knowledge maps (CKMs) store or predict radio channel quantities as functions of position and environment, enabling faster planning and adaptation than repeated ray tracing or exhaustive measurements [1], [2]. In UAV enabled 6G networks, CKMs are especially attractive because altitude changes both the geometry and the line of sight probability of the link [3], [4]. A practical CKM surrogate, however, must generalize to urban layouts not seen during training and must not hide hard propagation regimes behind a single global score.

The central difficulty is that the three targets do not behave like one smooth image regression problem. LoS attenuation has strong radial and height structure, NLoS attenuation depends on blocked link regimes, and delay and angular spread contain near zero LoS values with heavier NLoS tails. The contribution is HARP-NET CKM, the Height-Aware Residual Prior Network for Channel Knowledge Map prediction. It is a dense CKM recipe for one fixed UAV simulation contract: strict city holdout, valid receiver masking, explicit LoS/NLoS and height reporting, priors calibrated only on the training split, and a bounded neural correction around those priors.

A. Radio Map and Channel Parameter Predictors

Dense radio map prediction is often framed as image to image regression. RadioUNet [5], PMNet, and the ICASSP path loss challenge family showed that U-Net style models can be strong dense channel attenuation baselines [6]–[8]. Later work adds attention, equivariance, domain shift handling, corridor weighting, or digital twin generation [9]–[13]. These papers give the closest dense channel attenuation context, but not the same contract: continuous UAV transmitter height, strict city holdout on CKM samples, and three dense targets (channel attenuation/PL, DS, AS).

CKM work also emphasizes that a radio map is not merely an image but a position indexed channel side information object [1], [2], [14]. A random image split can place visually similar urban layouts in both training and testing; a city split tests transfer to unseen layouts, so the evaluation protocol is part of the method.

Physics informed and hybrid methods address a different weakness of pure image regression: the model should not need to rediscover stable propagation effects from pixels alone. ReVeal [15] uses physics informed constraints for radio environment mapping, while classical air to ground, COST/Hata, and two ray models encode known distance, height, reflection, and elevation angle structure [3], [16]–[21]. This paper uses those models only as shape priors; all deployment coefficients are calibrated from training cities under the same CKM masking convention used at inference.

The spread side has fewer direct dense map references. Yang et al. predict A2G delay spread with classical regressors [22]; Huang et al. use a Transformer style UAV mmWave predictor for received power, RMS delay spread, and RMS angular spread [23]; and Mi et al. predict mmWave channel attenuation (PL), DS, AS, and K-factor from measurement linked point clouds [24]. These works are close in target type but still predict link level or sparse channel parameters, not dense 513×513 CKM maps for unseen cities.

Finally, heteroscedastic regression [25], mixture density wireless predictors [26]–[28], and stochastic radio map models [29] all point to the same issue: NLoS and spread targets are not well represented by a single symmetric residual. For DS/AS, the closest external references are therefore context rather than direct CKM benchmarks.

II. DATASET AND TASK CONTRACT

Each sample is a 513×513 urban topology map from the ray traced CKM setting used in this work. The carrier is fixed at 7.125 GHz, so frequency is not a learned variable; the main transmitter degree of freedom is height. There is one UAV transmitter at the map centre. The transmitter horizontal position is fixed, but its antenna height h_{tx} varies by sample

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TABLE I
EXPERIMENTAL CONTRACT USED FOR ALL REPORTED FINAL NUMBERS.

Item	Setting
Map resolution	513×513, $1\,\mathrm{m}\times 1\,\mathrm{m}$ pixels
Carrier / receiver	7.125 GHz; $h_{\mathrm{rx}} = 1.5\,\mathrm{m}$
Transmitter	One centred UAV; stored $h_{\mathrm{tx}} \approx 12\,\mathrm{m}$ to $478\,\mathrm{m}$, nominal beta law 10 m to 500 m
Inputs	Topology, ground/visibility masks, frozen priors, scalar height code
Targets	Channel attenuation / PL (dB), DS (ns), AS (deg), one dense map per target
Metric mask	Non building ground receivers only; building pixels excluded
Split policy	City holdout; validation selects the checkpoint, test is final only
Samples	Train 10840; validation 2750; test 2590; all 16180

and is treated as a central conditioning variable rather than as metadata. The released CKM files call this node a UAV, but the model contract is an elevated 7.125 GHz transmitter within the sampled height range, so the same setup can also represent other elevated radio access nodes under the same simulation assumptions. The spatial resolution is $1\,\mathrm{m}\times 1\,\mathrm{m}$ per pixel. Every non building pixel is treated as an active receiver location, with the receiver antenna placed at $h_{\mathrm{rx}} = 1.5\,\mathrm{m}$ above the ground. Building pixels are excluded from the receiver set rather than treated as easy zero error targets. The model predicts three aligned dense maps:

$$y_t \in \{\mathrm{PL}\ (\mathrm{dB}), \mathrm{DS}\ (\mathrm{ns}), \mathrm{AS}\ (\mathrm{deg})\}.$$

The CKM HDF5 field and implementation call the first target `path_loss` and retain the abbreviation PL. In this paper, PL is interpreted as the dense channel attenuation map in dB, while *path loss* is kept for standard propagation formulas, cited path loss models, and dataset/code names that use that term. Let $m_i(p) = 1$ if pixel p of sample i is one of these valid ground receivers, and zero otherwise. For a split $\mathcal{S}$, the target specific pixel weighted RMSE is

$$\mathrm{RMSE}_t = \sqrt{\frac{\sum_{i\in\mathcal{S}}\sum_p m_i(p)\left(\hat{y}_{i,t}(p) - y_{i,t}(p)\right)^2}{\sum_{i\in\mathcal{S}}\sum_p m_i(p)}}. \tag{1}$$

The same receiver-mask formula is used for all three targets. For DS and AS, each valid ground pixel is still a receiver with a ray-traced ground truth spread value, so the reported spread RMSE is the dense prediction compared with the dense CKM target over all valid ground receivers.

The split is city based rather than random over pixels or tiles. This matters: random splits would allow the model to see nearly identical urban morphology during training and testing. The final held out test split contains 2590 samples from cities not used to optimize model weights. Validation is used for model selection; test is used only for the final generalization claim. Table II lists the exact city assignment. Unless explicitly stated otherwise, every final model result reported below refers to the separate 2590-sample, 14-city test split.

The transmitter height distribution is intentionally not uniform. Direct comparison against the stored HDF5 histogram shows that the closest tested affine version of the approximate double beta sampler is the original $h_{\mathrm{design}} \approx 10 + 490X$, with

$$X \sim 0.75\,\mathrm{Beta}(3, 23) + 0.25\,\mathrm{Beta}(4, 4),$$

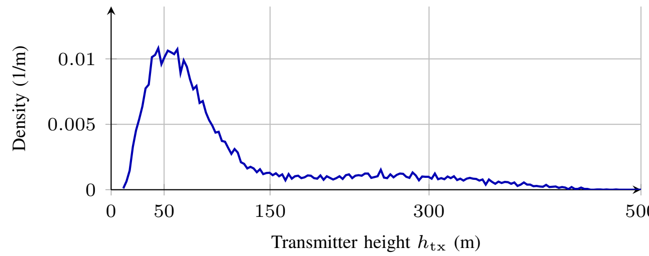

Fig. 1. Empirical transmitter height density read directly from the CKM HDF5 database using the 3 m histogram of (2), linearly interpolated between bin centres for display. Most training support lies in the low and middle altitude bands.

and nominal support 10 m to 500 m. Reported results use the empirical HDF5 realization: h_{tx} spans 11.61 m to 477.54 m, with mean 114.02 m and median 75.66 m. Heights above 478 m are possible under the nominal law but rare, so the observed maximum is consistent with finite sampling. Let $\Delta h = 3\,\mathrm{m}$, $B_k = [10 + 3k, 13 + 3k)$, and h_i be the stored height of sample i. The reporting histogram is

$$\hat{f}_{\Delta h}(h) = d_k, \quad h \in B_k,$$
$$d_k = \frac{n_k}{N\Delta h}, \quad n_k = \sum_{i=1}^{N} \mathbf{1}\{h_i \in B_k\}, \quad N = 16180. \tag{2}$$

For readability in Fig. 1, the plotted curve linearly interpolates between the bin centre values (c_k, d_k), with $c_k = 11.5 + 3k$. All database counts are 3972 samples below 50 m, 8456 from 50 m to 150 m, 2490 from 150 m to 300 m, and 1262 above 300 m; the corresponding training counts are 2676, 5654, 1644, and 866.

A. Quantized Targets and Continuous Outputs

The released CKM targets are not ideal real valued measurements. Channel attenuation is stored on an approximately 1 dB grid, while delay spread and angular spread are strongly integer valued and visibly staircase like in many maps. The model outputs are not quantized to these grids: the residual network emits continuous values, and all reported metrics compare those continuous outputs against the stored targets. This matters for interpretation: the final path loss MAE is 0.9298 dB, close to the label step, while RMSE remains 1.6519 dB because it emphasizes hard tails. Less quantized labels would likely expose more of the continuous surrogate's resolution, especially for small residual improvements and spread map boundaries.

III. HARP-NET CKM FRAMEWORK

A. Why the Final Method Is Hybrid

HARP-NET CKM separates stable structure from the residual structure that remains hard to model analytically. First, the attenuation prior captures LoS distance, height and two ray effects and adds a calibrated NLoS branch for blocked regions. Second, the spread priors use log domain regime calibration for delay spread and angular spread. Third, the neural model receives these frozen maps and predicts only bounded residual corrections. This division keeps the predictable part of the map

TABLE II
CITY HOLDOUT ASSIGNMENT USED BY THE DISTRIBUTION FIRST AND FINAL NEURAL MODELS.

Split	Samples	Cities	City names
Train	10840	37	Abidjan, Abu Dhabi, Alexandria, Antigua Guatemala, Athens, Bangalore, Bangkok, Bath, Berlin, Bratislava, Brugge, Buenos Aires, Canberra, Chefchaouen, Dalian, Hanoi, Ho Chi Minh City, Hyderabad, Kinshasa, Leuven, Ljubljana, Luanda, Lyon, Manaus, Minsk, Nagpur, Paris, Rio de Janeiro, Santiago, Shanghai, Siem Reap, St. Louis, Thane, Toledo, Valparaiso, Venice, Vienna.
Validation	2750	15	Bilbao, Chiang Mai, Cleveland, Dhaka, Dubrovnik, Fortaleza, Marrakesh, Mexico City, Munich, Nazareth, Nice, Salvador, Segovia, Tunis, Victoria.
Test	2590	14	Barcelona, Beijing, Carcassonne, Copenhagen, Halifax, Jaipur, Johannesburg, Key West, Kuala Lumpur, Osaka, Phuket, Pune, Surat, Vancouver.

TABLE III
INPUT/OUTPUT CONTRACT OF THE FULL FRAMEWORK AND FINAL NEURAL MODEL.

Stage	Variables
External framework inputs	Building height map and UAV transmitter height h_{tx}, stored as `uav_height` in the CKM files.
Computed support maps	LoS/NLoS visibility mask and valid ground receiver mask.
Frozen prior maps	Combined path loss prior, LoS path loss prior, NLoS path loss prior, delay spread prior, and angular spread prior.
Inputs to HARP-NET CKM	Topology map, transmitter height conditioning, ground mask, LoS mask, NLoS mask, and the five frozen prior maps.
Final outputs	Channel attenuation, delay spread, and angular spread maps.

auditable and reserves learning capacity for the local errors that the frozen estimators cannot explain.

In this paper, a *prior* means a frozen estimator fitted only on training cities, not a hand tuned curve evaluated on the test set. The prior sees the same support information available at deployment: centred transmitter geometry, transmitter height, topology derived morphology, and the geometric LoS/NLoS mask. Once its coefficients and fallback tables are stored, no gradient update is applied to it inside the neural training loop. This separation is important because it lets the final residual model be judged against a demanding baseline: the model must improve an already calibrated predictor rather than merely beat FSPL.

Thus the full generator has two external inputs and three final outputs. The neural residual model has nine spatial map channels plus a separate scalar transmitter height conditioning path because the framework has already computed the visibility mask, receiver mask, and frozen prior maps.

B. Frozen Priors

The path loss prior combines a coherent two ray LoS branch with a calibrated NLoS branch. More precisely, (4) is the Friis/free space dB term and (5) is the standard coherent direct plus ground reflected image source construction [16], [17], also used in low altitude A2G two ray modeling [21]. The fitted ρ, ϕ, b, and r curves are not taken from those references; they are CKM train split calibration terms wrapped around that two ray structure. The NLoS branch uses the COST231/Hata urban macro expression as a coarse blocked link trend [18], combines it with an Al-Hourani style elevation dependent A2G excess loss envelope [19], and keeps the LoS/NLoS regime separation used in UAV and standardized channel model literature [3], [20]. The spread priors use the log domain treatment of large scale delay/angle spread parameters from 3GPP/WINNER style models [20], [30]; the UAV empirical literature motivates the elevation and angular spread terms rather than supplying the CKM coefficients directly [31], [32]. These references define the shape assumptions; all numerical coefficients used here are still fitted on the training cities only. The UAV remains horizontally centred, so d_{2D} is the per pixel distance from the centre transmitter to each receiver. The height h_{tx}, however, changes from sample to sample and is one of the main physical degrees of freedom in the method. For receiver height h_{rx} and carrier frequency f_{MHz}, the direct and reflected distances are

$$\begin{aligned} d_{\mathrm{LoS}} &= \sqrt{d_{2D}^2 + (h_{\mathrm{tx}} - h_{\mathrm{rx}})^2}, \\ d_{\mathrm{ref}} &= \sqrt{d_{2D}^2 + (h_{\mathrm{tx}} + h_{\mathrm{rx}})^2}. \end{aligned} \tag{3}$$

For the positive UAV and receiver heights used here, d_{ref} is strictly larger than d_{LoS}; equality would occur only in the limiting geometric case where one antenna lies exactly on the ground plane. The LoS branch starts from

$$\mathrm{FSPL} = 32.45 + 20\log_{10}\left(\frac{d_{\mathrm{LoS}}}{1000}\right) + 20\log_{10}(f_{\mathrm{MHz}}) \tag{4}$$

and adds an effective coherent reflection correction

$$C_{2\mathrm{ray}} = -20\log_{10}\left|1 + \rho(h_{\mathrm{tx}})\frac{d_{\mathrm{LoS}}}{d_{\mathrm{ref}}}e^{-j(k(d_{\mathrm{ref}}-d_{\mathrm{LoS}})+\phi(h_{\mathrm{tx}}))}\right|. \tag{5}$$

The final LoS prior is

$$\mathrm{PL}_{\mathrm{LoS}} = \mathrm{FSPL} + C_{2\mathrm{ray}} + b(h_{\mathrm{tx}}) + r(d_{2D}, h_{\mathrm{tx}}), \tag{6}$$

where b is a height dependent bias and r is a smoothed radial residual fit. The direct distance d_{LoS} sets the free space baseline, d_{ref} supplies the image source ground reflected path, and $k(d_{\mathrm{ref}} - d_{\mathrm{LoS}})$ is the reflected wave phase lag. The calibrated $\rho(h_{\mathrm{tx}})$, $\phi(h_{\mathrm{tx}})$, $b(h_{\mathrm{tx}})$, and $r(d_{2D}, h_{\mathrm{tx}})$ curves are fitted in 5 m height bins, smoothed, interpolated at inference, and guarded to the $[20, 180]$ dB range. The notation is local: $\rho(h_{\mathrm{tx}})$ is the LoS reflection amplitude, whereas ρ_k denotes local building density; similarly $b(h_{\mathrm{tx}})$ is a LoS bias, while b_q^{nat}, b_{top}, and b_{41} belong to the spread prior.

The LoS calibration is deliberately structured rather than fully flexible. It is fitted only on valid LoS ground pixels from training cities. First, the effective reflection amplitude ρ and phase ϕ are fitted in 5 m transmitter height bins to reproduce the ring like constructive and destructive interference visible in LoS channel attenuation maps. Second, the per height bias b removes a remaining vertical offset after the coherent two ray fit. Third, the radial residual table $r(d_{2D}, h_{\mathrm{tx}})$ averages the remaining error by distance from the centred transmitter, smooths it across radius, clips it to a small dB range, and stores it by height. At inference, the actual h_{tx} selects/interpolates

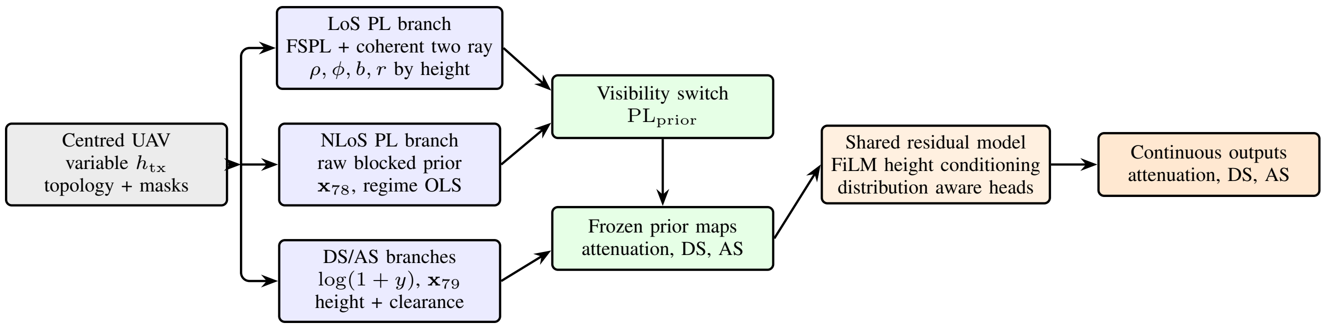

Fig. 2. Compact HARP-NET CKM pipeline. The transmitter is always centred, but its height varies by sample. Frozen height aware priors are produced first and then used as channels for the final HARP-NET CKM model.

these calibration curves, so the LoS prior changes continuously with antenna height even though the UAV horizontal coordinate remains fixed at the map centre.

This explicit LoS branch is diagnostic: many CKM LoS errors are radial rings around the centred transmitter rather than building edge artifacts. The two ray plus radial table removes that height/radius component before the network spends capacity on local visibility and morphology corrections.

The NLoS prior starts from the maximum of COST231 like and A2G-NLoS raw terms. The first term below is the COST231/Hata urban macro formula adapted as a shape prior for the CKM geometry, not as an assertion that the original model is directly valid at every transmitter height:

$$\begin{aligned} \mathrm{PL}_C &= 46.3 + 33.9\log_{10}(f_{\text{MHz}}) \\ &\quad - 13.82\log_{10}(h_{\text{tx}}) - a(h_{\text{rx}}) \\ &\quad + \eta(h_{\text{tx}})\log_{10}(d_{\text{km}}) + 3, \\ \eta(h_{\text{tx}}) &= 44.9 - 6.55\log_{10}(h_{\text{tx}}), \end{aligned} \tag{7}$$

where C denotes the COST231-shaped branch, $d_{\text{km}} = \max(d_{2D}/1000, 0.001)$, and $a(h_{\text{rx}}) = (1.1\log_{10} f_{\text{MHz}} - 0.7)h_{\text{rx}} - (1.56\log_{10} f_{\text{MHz}} - 0.8)$ is the small/medium city mobile height correction. The $+3$ dB term is retained as the COST231 metropolitan correction inside this empirical basis; it is not a claim that the original COST231 deployment domain transfers unchanged to 7.125 GHz UAV links. The second term is a conservative A2G NLoS envelope: it keeps altitude/elevation structure from coverage models, then lets the train only calibration correct the fixed excess loss constants:

$$\begin{aligned} \mathrm{PL}_{\text{A2G,NLoS}} &= \lambda_0(h_{\text{tx}}, f) + \beta_{\text{NLoS}} \\ &\quad + A_{\text{NLoS}}\exp\left(-\frac{90-\theta^\circ}{\tau_{\text{NLoS}}}\right), \end{aligned} \tag{8}$$

with $\lambda_0 = 20\log_{10}(4\pi h_{\text{tx}} f/c)$, $\beta_{\text{NLoS}} = -16.16$ dB, $A_{\text{NLoS}} = 12.0436$, and $\tau_{\text{NLoS}} = 7.52^\circ$. The raw blocked link prior is the conservative envelope

$$P_0 = \mathrm{PL}_{\text{NLoS,raw}} = \max(\mathrm{PL}_C, \mathrm{PL}_{\text{A2G,NLoS}}). \tag{9}$$

The maximum is deliberately pessimistic: whichever raw branch predicts stronger attenuation is kept. It is then corrected by train only regime wise linear calibration:

$$\mathrm{PL}_{\text{NLoS}} = \mathrm{clip}(\mathbf{x}_{78}^\top \mathbf{w}_{78,r}, 20, 180). \tag{10}$$

The path loss prior is therefore not one formula with a small correction. It is two different calibrated feature extractors. The LoS side calibrates height binned two ray amplitude/phase, height bias, and radial residual curves. The NLoS side adds calibration terms for height, building topology, and density. This feature choice follows the same modelling instinct as COST 231 Walfisch Ikegami: NLoS urban loss should depend on the surrounding built form, not only on range and frequency [18]. The Walfisch Bertoni component treats rows of buildings as repeated diffraction screens [33]. The Ikegami street level component motivates descriptors such as building height, street width, street orientation, and mobile height [34]. In the CKM maps these coarse planning variables are replaced by directly observed local windows: building density, mean building height, and NLoS support fractions. A modern machine learning analogue is Juang's path profile work, where building profile statistics are used in a linear model to explain urban path loss predictions [35]. The raw blocked link prior is therefore only one feature in a 15 term morphology vector:

$$\begin{aligned} \mathbf{x}_{78} = [&P_0^2, P_0, \log(1+d_{2D}), \rho_{15}, \rho_{41}, \\ &h_{15}, h_{41}, \rho_{41}\log(1+d_{2D}), n_{15}, \\ &n_{41}, n_{41}\log(1+d_{2D}), \sigma_{\text{shadow}}, \theta_{\text{norm}}, \\ &n_{41}\theta_{\text{norm}}, 1]^\top, \end{aligned} \tag{11}$$

where P_0 is the raw NLoS prior, ρ_{15} and ρ_{41} are the fractions of building pixels in 15×15 and 41×41 windows, h_{15} and h_{41} are local building height summaries, n_{15} and n_{41} summarize local NLoS support, σ_{shadow} is an elevation dependent shadowing proxy, and θ_{norm} is the normalized elevation angle feature. The interaction $\rho_{41}\log(1+d_{2D})$ lets density matter differently with range, $n_{41}\log(1+d_{2D})$ lets local blockage support vary with range, and $n_{41}\theta_{\text{norm}}$ activates when obstruction and elevation jointly matter. The constant 1 is the regime wise bias term. The deployed path loss prior is selected by the geometric visibility mask:

$$\mathrm{PL}_{\text{prior}} = m_{\text{LoS}}\mathrm{PL}_{\text{LoS}} + (1 - m_{\text{LoS}})\mathrm{PL}_{\text{NLoS}}. \tag{12}$$

The regime index r is not a learned latent class. It is selected from observable scene descriptors: topology/morphology class, LoS/NLoS state, and antenna height group, with fallback calibrations when an exact regime has too few training pixels. Thus the prior is auditable: one can inspect the raw blocked estimate P_0, the local morphology terms, the selected regime, and the final linear correction separately.

This NLoS branch is intentionally less physical than the LoS branch. The visibility mask says that the direct ray is blocked, but it does not say which street canyon, rooftop edge, or local clutter feature dominates the strongest remaining path. The calibrated feature vector therefore treats COST/A2G values as one input among morphology summaries. It is a compromise between a fully analytic blocked link law, which was too rigid, and a deep model, which was less transparent when trained from scratch.

Table IV makes this provenance explicit. The only pieces taken directly as literature structures are the Friis/two ray geometry, the COST231/Hata urban trend, the A2G elevation loss shape, and the log domain spread convention. The reflection curves, radial residuals, OLS/ridge weights, fallback hierarchy, and morphology windows are fitted or computed inside this CKM dataset pipeline.

The LoS two ray parameters are fitted by a small grid search rather than by a neural optimizer. For each 5 m height bin B_h, candidate values $\hat{\rho} \in \{0, 0.05, \ldots, 1.5\}$ and $\hat{\phi} \in [-\pi, \pi)$ on a 48-point phase grid are evaluated on valid LoS training pixels. The upper end of this grid is a robustness allowance for an effective CKM fitted coefficient, not a claim that a passive Fresnel reflection coefficient exceeds unity; the smoothed deployed reflection amplitude curve stored by the final prior remains below unity. For each candidate pair, the dB bias has the closed form value

$$\hat{b}(\hat{\rho}, \hat{\phi}) = \frac{1}{|B_h|} \sum_{i \in B_h} \left[y_i - \mathrm{FSPL}_i - C_{2\mathrm{ray},i}(\hat{\rho}, \hat{\phi}) \right], \tag{13}$$

and the selected pair minimizes the corresponding RMSE. The fitted ρ, ϕ, and b curves are then smoothed across neighbouring height bins, weighted by bin support, and linearly interpolated at inference.

The NLoS branch uses the same train only principle but with a linear morphology calibrator. For every supported regime r, the calibration weights solve

$$\hat{\mathbf{w}}_{78,r} = \arg\min_{\mathbf{w}} \sum_{i \in r} \left(\mathbf{x}_{78,i}^{\top} \mathbf{w} - y_i \right)^2, \tag{14}$$

with fallback to stored antenna height keys within the same morphology and LoS/NLoS group when the exact antenna height key is unavailable. At deployment, the same deterministic features select one stored coefficient vector and produce the clipped NLoS prior in (10). This is an ordinary least squares objective, not a requirement that every regime have a nonsingular normal matrix; if a calibration block is rank deficient or ill conditioned, the stored solution is interpreted as the numerical least squares or pseudoinverse solution associated with this objective.

The delay spread and angular spread priors are fitted in a transformed domain,

$$z = \log(1 + y), \qquad \hat{y} = \exp(\hat{z}) - 1, \tag{15}$$

because both standard channel models and the CKM histograms make spread quantities more regular in log space [20], [30]. The released spread maps have a near zero mass plus a long tail: LoS direct path pixels cluster near zero, while blocked or rooftop scattered pixels create rare but important large values. The standardized models justify the log domain large scale variable; the particular range, elevation, density, height, and NLoS support terms below are CKM specific deterministic covariates chosen to make that log variable depend on observable map geometry. Their raw prior is

$$\begin{aligned} z^{(0)} = \; & \log(1 + b_q^{\mathrm{nat}}) + b_{\mathrm{top}} + a_1 \log(1 + d_{2D}) + a_2 \theta_{\mathrm{inv}} \\ & + a_3 \rho_{41} + a_4 h_{41} + a_5 n_{41} + a_6 n_{41} \theta_{\mathrm{inv}}, \end{aligned} \tag{16}$$

The constants in (16) are fixed before ridge calibration. For delay spread, $b_{\mathrm{LoS}}^{\mathrm{nat}} = 4.0\,\mathrm{ns}$, $b_{\mathrm{NLoS}}^{\mathrm{nat}} = 11.0\,\mathrm{ns}$, and $(a_1, \ldots, a_6) = (0.11, 0.62, 0.48, 0.32, 0.86, 0.60)$. For angular spread, $b_{\mathrm{LoS}}^{\mathrm{nat}} = 3.0^\circ$, $b_{\mathrm{NLoS}}^{\mathrm{nat}} = 7.0^\circ$, and $(a_1, \ldots, a_6) = (0.05, 0.52, 0.60, 0.38, 0.58, 0.35)$. The topology offsets b_{top} are log domain offsets, not native ns/degree offsets. In the class order open sparse low rise, open sparse vertical, mixed compact low rise, mixed compact mid rise, dense block mid rise, and dense block high rise, the delay spread offsets are $(0.00, 0.18, 0.06, 0.22, 0.24, 0.34)$, and the angular spread offsets are $(0.00, 0.09, 0.05, 0.16, 0.18, 0.24)$. The six topology classes are assigned from map level building coverage $\bar{\rho}_{\mathrm{map}}$ and mean above ground building height $\bar{h}_{\mathrm{map}}$: coverage $\bar{\rho}_{\mathrm{map}} \le 0.12$ is open sparse, with low rise if $\bar{h}_{\mathrm{map}} \le 12\,\mathrm{m}$ and vertical otherwise; coverage $\bar{\rho}_{\mathrm{map}} \ge 0.28$ is dense block, with mid rise if $\bar{h}_{\mathrm{map}} \le 28\,\mathrm{m}$ and high rise otherwise; intermediate coverage is mixed compact, with low rise if $\bar{h}_{\mathrm{map}} \le 12\,\mathrm{m}$ and mid rise otherwise. These labels are map level routing keys for calibration, not per pixel learned features.

Here q is the LoS/NLoS state. The native anchor b_q^{nat} gives a different starting scale to LoS and NLoS pixels, b_{top} shifts the baseline by topology class, $\log(1 + d_{2D})$ captures range growth, θ_{inv} increases the prior for low elevation links, and ρ_{41}, h_{41}, and n_{41} summarize local building density, height, and NLoS support. This raw log prior is intentionally simple; it encodes the expected spike plus tail shape before CKM specific calibration. A 23-term feature vector then augments $z^{(0)}$ with range, elevation, height, multi scale morphology, blocker depth, clearance, and interaction terms:

$$\begin{aligned} \mathbf{x}_{79} = [& (z^{(0)})^2, z^{(0)}, \log(1 + d_{2D}), \theta_{\mathrm{norm}}, \theta_{\mathrm{inv}}, h, h^2, \\ & \rho_{15}, \rho_{41}, h_{15}, h_{41}, n_{15}, n_{41}, n_{41} \log(1 + d_{2D}), \\ & \rho_{41} \theta_{\mathrm{inv}}, b_{41}, 1, c_{41}, t_{41}, \theta_{\mathrm{norm}} \rho_{41}, \\ & c_{41} \theta_{\mathrm{inv}}, t_{41} \rho_{41}, t_{41} n_{41}]^{\top}. \end{aligned} \tag{17}$$

Here h is normalized transmitter height, b_{41} is a local blocker depth summary, c_{41} is transmitter clearance over nearby buildings, and t_{41} is the fraction of locally taller buildings. These height terms are not incidental: they tell the spread prior whether the centred UAV is above the local skyline, near rooftop level, or below nearby high rise clutter. The quadratic raw prior terms let the calibration bend the initial log domain estimate, while the interaction terms represent cases where range, low elevation, dense buildings, and NLoS support become harmful together; in particular, $c_{41} \theta_{\mathrm{inv}}$ means that transmitter clearance is allowed to matter differently at low elevation, not that clearance is only important when NLoS support is already high. The calibrated log prior is ridge regularized per metric and regime:

$$\hat{\mathbf{w}}_{79,r} = \left(\mathbf{X}^{\top} \mathbf{X} + \lambda \mathbf{D}^{\top} \mathbf{D} \right)^{-1} \mathbf{X}^{\top} \mathbf{z}. \tag{18}$$

TABLE IV
TERM BY TERM PROVENANCE OF THE MAIN FROZEN PRIOR QUANTITIES.

Term	Prior block	Source	Role in this paper
Literature formula structures			
d_{LoS}	LoS PL	CKM geometry + Friis distance term [16], [17]	Direct transmitter–receiver distance used in FSPL.
d_{ref}	LoS PL	Image source two ray geometry [16], [17], [21]	Length of the single ground reflected ray.
$C_{2\mathrm{ray}}$	LoS PL	Coherent direct/reflected field sum [16], [17]	Constructive/destructive radial rings before learned correction.
P_0	NLoS PL	COST231/Hata trend [18] + A2G elevation envelope [3], [19]	Raw blocked link scale before morphology OLS calibration.
CKM deterministic geometry and morphology features			
ρ_k, h_k, n_k	NLoS PL and spreads	Topology and LoS/NLoS masks	Local density, building height, and NLoS support in a $k \times k$ window.
b_{41}, c_{41}, t_{41}	Spreads	Obstruction features motivated by A2G clutter behavior [3], [31], [32]	Blocker depth, transmitter clearance, and taller building fraction.
$\theta_{\mathrm{norm}}, \theta_{\mathrm{inv}}$	Spreads	Geometry + A2G elevation dependence [3], [19], [20]	Elevation angle and low elevation complement for blocked/spread prone regimes.
CKM train split fitted calibration			
$\rho(h_{\mathrm{tx}}), \phi(h_{\mathrm{tx}})$	LoS PL	Train split fit + smoothing	Smoothed deployed effective reflection amplitude and phase; not material constants.
$b(h_{\mathrm{tx}})$	LoS PL	Train split fit + smoothing	Smoothed deployed height dependent offset left after coherent two ray fitting.
$r(d_{2D}, h_{\mathrm{tx}})$	LoS PL	Train split fit + smoothing	Smoothed deployed radial residual left by two ray plus bias.
$b_q^{\mathrm{nat}}, b_{\mathrm{top}}, a_j$	Spreads	Fixed CKM raw prior constants	Initial log domain DS/AS scale before ridge calibration.

TABLE V
FROZEN PRIOR ESTIMATORS USED BY THE FINAL MODEL. ALL COEFFICIENTS ARE FITTED ON TRAINING CITIES AND FROZEN BEFORE NEURAL RESIDUAL TRAINING; CITATIONS INDICATE THE LITERATURE BASIS OF EACH STARTING STRUCTURE.

Target/regime	Starting structure	Train only calibration	Deployed prior map
PL LoS	FSPL plus coherent direct/reflected two ray correction [16], [17], [21]	$\rho(h_{\mathrm{tx}})$, $\phi(h_{\mathrm{tx}})$, $b(h_{\mathrm{tx}})$, and radial residual $r(d_{2D}, h_{\mathrm{tx}})$ in smoothed 5 m height bins	LoS channel attenuation map carrying the dominant height/radius rings before any neural residual is added.
PL NLoS	Conservative envelope of COST231-shaped urban loss and an elevation aware A2G-NLoS term [3], [18]–[20]	Regime wise OLS over $\mathbf{x}_{78}$, selected by morphology, visibility, and antenna height group with fallback keys and output clipping	NLoS channel attenuation map that stores blocked link scale, local density, blocker, and elevation effects.
DS / AS	Log domain raw spike plus tail scale from visibility, range, elevation, and local morphology [20], [30]–[32]	Ridge calibration over $\mathbf{x}_{79}$ per metric/regime, with height collapsed, topology wide, visibility wide, and global fallbacks	Delay- and angular spread maps in native units after $\exp(\hat{z}) - 1$, used as frozen statistical anchors.

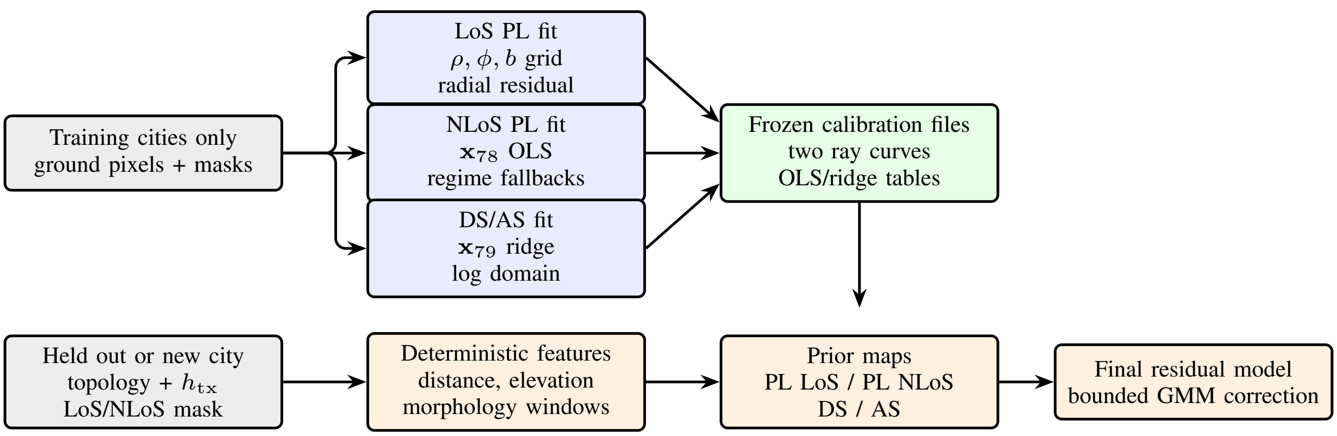

Fig. 3. Train only prior calibration and deployment. The upper row fits and stores the prior parameters using training cities only. The lower row applies those frozen tables to a held out sample before the neural residual model is called.

The calibrated log prior is then $\hat{z} = \mathbf{x}_{79}^{\top}\hat{\mathbf{w}}_{79,r}$. The diagonal regularizer $\mathbf{D}$ keeps the calibrated curve close to the raw prior unless the training residuals consistently justify a correction. In practice this matters for the spread targets: a flexible unregularized fit can chase rare high spread pixels, whereas the ridge fit keeps the near zero LoS bulk and the NLoS tail connected to the same physical features.

The spread prior should be read as a statistical prior rather than as a ray by ray multipath model. It does not attempt to reconstruct individual reflections. Instead, it predicts the expected scale of a dense spread map from height, range, LoS/NLoS state, and local morphology. The log transform makes the near zero mass and the long tail share one regression space, while the regime fallbacks prevent small calibration subsets from inventing unsupported high spread behaviour.

All calibrations use training cities only. The spread prior dictionary contains 114 fitted regime keys: 36 exact topology/visibility/height keys per metric plus topology, visibility, and global fallbacks. Height is handled differently across priors: the LoS PL branch interpolates 5 m curves, whereas NLoS PL and spread branches use antenna height regimes (`low_ant`, `mid_ant`, `high_ant`) while still keeping continuous height, clearance, and taller building features inside $\mathbf{x}_{79}$. The point of this heavy calibration is to store many

TABLE VI
FROZEN PRIOR ONLY RMSE ON THE FINAL HELD OUT TEST SPLIT. THESE ARE THE ANCHORS THAT THE RESIDUAL MODEL MUST IMPROVE.

Target	Overall	LoS	NLoS	Unit
Attenuation	1.9383	1.7519	3.5143	dB
Delay spread	28.1023	27.4851	29.6157	ns
Angular spread	13.7416	15.2818	8.6614	deg

small decisions about morphology, visibility, and height before the neural residual is trained. On the final held out test split, the deployed prior anchors are summarized in Table VI. Beyond the topology, masks, and scalar height path, the final neural model receives the deployed prior maps rather than the handcrafted feature vectors; those features remain inside the frozen prior estimators instead of becoming additional learned inputs.

C. HARP-Net CKM Residual Model

The neural model receives nine aligned channels: topology, LoS mask, NLoS mask, ground mask, combined path loss prior, LoS path loss prior, NLoS path loss prior, delay spread prior, and angular spread prior. Height is encoded with sinusoidal features and injected through FiLM modulation [36], so the final network can adapt the residual correction continuously between the calibrated height regimes instead of learning one height agnostic correction. The backbone is a shared U-Net style encoder decoder with GroupNorm, FiLM conditioning, and task specific residual heads.

The actual final model is therefore not a replacement for the priors. Its input tensor already contains the deployed attenuation and spread estimates, and its output head is constrained to make bounded corrections around them. This choice changes the learning problem: the encoder decoder does not need to infer absolute free space attenuation, altitude trends, or the average spread scale from pixels alone. It learns where the frozen estimates are locally biased and how the remaining residual distribution should be placed across the map.

The height path mirrors the full implementation. The scalar h_{tx} is expanded to a 32-dimensional sinusoidal code using 16 log spaced frequencies over the transmitter height range,

$$\mathbf{e}(h_{\mathrm{tx}}) = \left[\sin\frac{h_{\mathrm{tx}}}{f_i}, \cos\frac{h_{\mathrm{tx}}}{f_i}\right]_{i=1}^{16} \in \mathbb{R}^{32}, \tag{19}$$

then mapped by an MLP to a $\mathbb{R}^{160}$ condition vector. Each FiLM site projects this vector to channel wise scale and shift parameters and modulates a feature map as

$$\mathbf{X}' = \mathbf{X} \odot (1 + \mathbf{g}) + \mathbf{b}. \tag{20}$$

Encoder FiLM uses the height condition directly; decoder FiLM and the global task heads use a fused condition formed from the height code and the pooled bottleneck context. Thus the same centred topology can receive different residual corrections when the UAV is at 30 m, 150 m, or 350 m. The height aware results in Sec. V show that this is not only an architectural convenience: sinusoidal height embeddings with FiLM modulation are sufficient for one model to cover the full varying height range without training a separate network per altitude bin.

The head uses a two stage residual logic around the frozen prior. The global stage asks: "what residual distribution does this whole map, height, and topology imply?" The local stage asks: "where should each residual component be placed in the 513×513 map?" This is the key distinction from a direct U-Net regressor. The network does not directly overwrite the prior with a free form map; it predicts a small residual distribution and then assigns that residual pixel by pixel.

For each task k, visibility region $r \in \{\mathrm{LoS}, \mathrm{NLoS}\}$, component $j \in \{1,2,3\}$, and pixel p, the global head receives the pooled bottleneck and fused height/context vector and outputs

$$\{\pi_{k,r,j}^{(g)}, \delta_{k,r,j}^{(g)}, \sigma_{k,r,j}^{(g)}\}_{j=1}^{3}.$$

After the implementation transforms, $\pi^{(g)}$ is a map level residual mixture distribution, $\delta^{(g)}$ is a bounded global residual offset, and $\sigma^{(g)}$ is a bounded global uncertainty scale. The local head receives the full resolution decoder feature map and outputs

$$\{p_{k,r,j}(p)\}_{j=1}^{3}, \quad \delta_{k,r}^{(\ell)}(p), \quad \alpha_{k,r}(p), \quad \tilde{\sigma}_{k,r}(p).$$

Here $p_{k,r,j}(p)$ is the per pixel component assignment, the local residual $\delta^{(\ell)}$ shifts all components at that pixel, α is an anchor gate initialized with bias -2 so the model starts close to the prior, and $\tilde{\sigma}$ is the local uncertainty term. This extends the distribution aware line of work from scalar mixture predictors to dense residual maps [26]–[28]. The native residual is clipped by target and visibility:

$$\Delta_{k,r,j}(p) = \mathrm{clip}\Big(\delta_{k,r,j}^{(g)} + \delta_{k,r}^{(\ell)}(p), -\delta_{\max,k,r}, \delta_{\max,k,r}\Big), \tag{21}$$

with caps of 7/25 dB for attenuation (PL) LoS/NLoS, 20/20 ns for DS, and 17°/15° for AS. The residual cap is specified in native units: dB for attenuation (PL), ns for DS, and degrees for AS. Omitting indices for compactness, the component mean is anchored to the frozen prior after this native residual has been assembled:

$$\begin{aligned} u^{\mathrm{nat}} &= y^{\mathrm{prior,nat}} + \alpha\Delta, \\ \mu &= u^{\mathrm{nat}}, \qquad k = \mathrm{PL}, \\ \mu &= \log(1 + [u^{\mathrm{nat}}]_+), \qquad k \in \{\mathrm{DS}, \mathrm{AS}\}. \end{aligned} \tag{22}$$

where $[u^{\mathrm{nat}}]_+ = \max(u^{\mathrm{nat}}, 0)$ and $0 \le \alpha_{k,r}(p) \le 1$. The attenuation branch (PL) is modeled in native dB space, while DS and AS use the log transform in (15). The deterministic map used for RMSE is the mixture mean in the modelled space:

$$\hat{z}_{k,r}(p) = \sum_{j=1}^{3} p_{k,r,j}(p)\mu_{k,r,j}(p). \tag{23}$$

For the attenuation branch (PL), $\hat{y}_{k,r} = \hat{z}_{k,r}$. For DS and AS, the output is mapped back with $\hat{y}_{k,r} = \exp(\hat{z}_{k,r}) - 1$. Thus the spread output is the deterministic point map used for RMSE, not the exact native space expectation of the log space mixture. The final map selects the LoS or NLoS branch with the explicit geometric mask. The mixture variances combine map level and local uncertainty,

$$\sigma_{k,r,j}(p) = \sqrt{\left(\sigma_{k,r,j}^{(g)}\right)^2 + (\tilde{\sigma}_{k,r}(p))^2}, \tag{24}$$

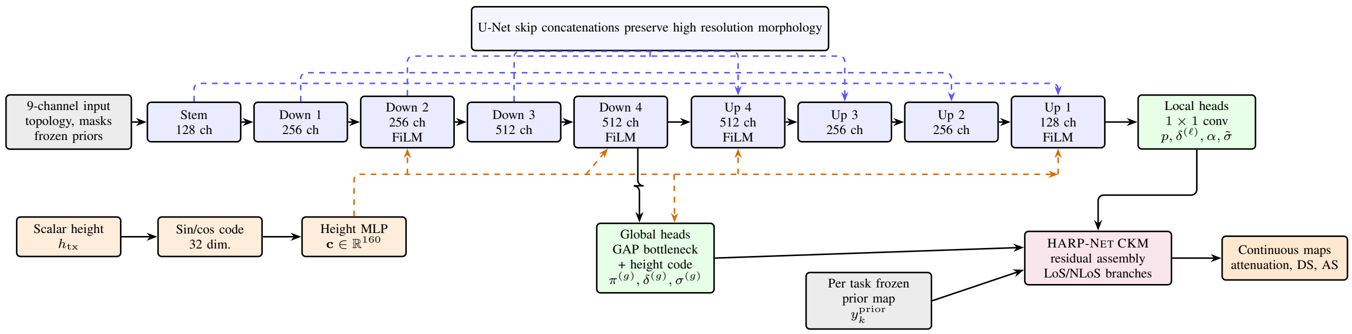

Fig. 4. Final shared deep learning architecture used in the reported system diagram. The map stack carries topology, masks, and frozen priors; the scalar Transmitter height is encoded separately and injected through FiLM. Encoder FiLM uses the height code; decoder FiLM and global heads use the fused height/bottleneck context. A shared U-Net produces local residual fields, pooled bottleneck features produce global GMM parameters, and the HARP-NET CKM residual assembly emits continuous channel attenuation, DS, and AS maps. The blue dashed arrows show the four encoder–decoder concatenations used in the implementation.

with both terms constrained to $[0.05, 3.0]$. During the probabilistic residual stage these variances and mixture weights define a pixel wise likelihood; during the final fine tuning stage the same head remains in place but the reported output is the deterministic point map above.

The two mixture weight objects have different jobs. The global weights $\pi^{(g)}_{k,r,j}$ describe the sample level residual histogram and are used by the distributional regularizer. The local weights $p_{k,r,j}(p)$ are the pixel wise assignment probabilities used in the final dense map. During the probabilistic residual stage the per pixel density is

$$q_{k,r}(z_k(p) \mid p) = \sum_{j=1}^{3} p_{k,r,j}(p)\,\mathcal{N}\big(z_k(p); \mu_{k,r,j}(p), \sigma^2_{k,r,j}(p)\big)\,, \tag{25}$$

and the regional NLL is the masked average of $-\log q_{k,r}(z_k(p) \mid p)$ over valid LoS or NLoS pixels. Separately, the KL term bins the true residuals $\epsilon_k(p) = y_k(p) - y_k^{\text{prior}}(p)$ into a soft empirical histogram and compares it with the histogram implied by $\{\pi^{(g)}_{k,r,j}, \delta^{(g)}_{k,r,j}\}$. This is what makes the head "distribution first": before the decoder places values pixel by pixel, the global branch is pressured to put mixture mass where the map residuals actually live.

This head is useful even when the final reported number is a deterministic RMSE. The probabilistic stage teaches the network that a hard NLoS region can contain several plausible residual levels, while the anchor gate decides how far the prediction may move from the frozen estimate. The final point map is therefore not an unconstrained average of arbitrary map values; it is the model space mean of a small residual mixture whose components remain tied to the prior.

HARP-NET CKM should not be read as only the final fine tuning loss. It has two active training phases: a large capacity probabilistic residual stage with probabilistic losses enabled, followed by an RMSE/MAE dominant fine tuning stage. The composite objective implemented by the model is

$$\begin{aligned}\mathcal{L} = \sum_k w_k \big(&\lambda_{\text{NLL}}\mathcal{L}_{\text{NLL},k} + \lambda_{\text{KL}}\mathcal{L}_{\text{KL},k} + \lambda_{\text{mom}}\mathcal{L}_{\text{mom},k} \\ &+ \lambda_{\text{anchor}}\mathcal{L}_{\text{anchor},k} + \lambda_{\text{guard}}\mathcal{L}_{\text{guard},k} + \lambda_{\text{out}}\mathcal{L}_{\text{out},k} \\ &+ \lambda_{\text{RMSE}}\mathcal{L}_{\text{RMSE},k} + \lambda_{\text{MAE}}\mathcal{L}_{\text{MAE},k}\big).\end{aligned} \tag{26}$$

The probabilistic terms matter during the probabilistic residual stage. The NLL term teaches the head to place mixture mass around the transformed target, the KL and moment terms keep the mixture distribution from becoming a numerically convenient but physically uninformative density, and the outlier budget term discourages a solution that spends most of its capacity on the easy bulk while leaving large tails unconstrained. The final fine tuning stage then disables those terms and optimizes the deterministic mixture mean point map for RMSE/MAE, with a small anchor and guard that keep the correction close to the prior. Table VII summarizes the roles. The selected checkpoint is therefore reported as a deterministic RMSE improvement over the frozen priors, not as a calibrated probabilistic predictor. In this final stage, checkpoint selection uses the validation composite $\text{PL}_{\text{RMSE}} + 0.30\,\text{DS}_{\text{RMSE}} + 0.25\,\text{AS}_{\text{RMSE}}$, matching the reported `score_weights`.

The practical role of the two stages is simple. The first stage makes the head behave like a residual density model: it learns where residual mass lies and discourages the old single mode collapse. The final fine tuning stage keeps the same architecture but shifts the optimization weight toward the deterministic metrics used for the paper tables. The anchor and guard terms remain active at small weight so the fine tuned model cannot gain average RMSE by damaging large regions where the frozen prior was already accurate.

IV. Experiments and Results

A. Main Test Results by Target

Table X reports the final unseen city test metrics. The model reduces RMSE relative to the frozen prior for all three targets. Channel attenuation is the cleanest success case because both RMSE and MAE improve. Angular spread also improves strongly in RMSE and remains approximately neutral in MAE. Delay spread improves RMSE while MAE slightly increases;

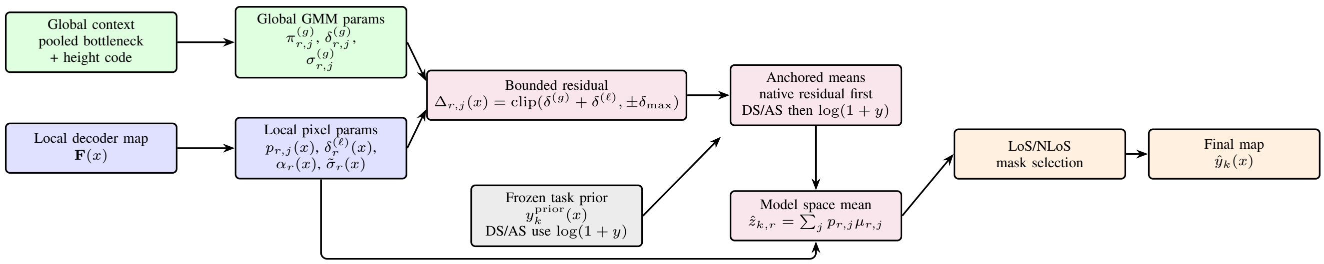

Fig. 5. HARP-NET CKM residual head. The global branch predicts the sample level residual mixture; the local branch places that distribution pixel by pixel through component weights, residual refinements, gates and local uncertainty. Component means are anchored to the frozen prior and the deterministic map is obtained from the pixel wise model space mixture mean after LoS/NLoS selection.

TABLE VII
LOSS TERMS AVAILABLE IN THE HARP-NET CKM OBJECTIVE AND THEIR PRACTICAL PURPOSE.

Term	Active stage	Role
GMM NLL	probabilistic residual stage	Fits the per pixel mixture density in native attenuation (PL) space and log transformed spread space.
Distribution KL	probabilistic residual stage	Regularizes the learned mixture so it remains close to the intended distributional structure instead of using degenerate weights or variances.
Moment matching	probabilistic residual stage	Encourages the model space mean and spread of the mixture to agree with target moments, stabilizing deterministic extraction from the GMM.
Outlier budget	probabilistic residual stage	Keeps rare large residuals visible in the objective instead of letting the global average hide them.
Anchor	both stages	Penalizes unnecessary deviation from the frozen prior; this is especially important because the prior is already a strong calibrated predictor.
Prior guard	both stages	Adds a penalty when the model becomes worse than the prior at valid pixels, discouraging destructive corrections.
RMSE / MAE	both stages, dominant in final fine tuning	Optimizes the deterministic map reported in the final evaluation; RMSE handles tails and MAE tracks typical absolute error.

TABLE VIII
TRAINING CONFIGURATION OF THE FINAL LINE.

Item	Value
Image size	513×513
Input channels	9 map channels plus height FiLM conditioning
Backbone	GroupNorm U-Net, base width 128, FiLM dim. 160
Residual head	3 mixture components, bounded residuals
Optimizer	AdamW, learning rate 5×10^{-5}, weight decay 0.01
Batching	batch size 1, gradient accumulation 8
Scheduler	1 warmup epoch plus cosine `LambdaLR`; early stopping patience 10
Augmentation	Training D4 rotations/reflections only; evaluation uses no augmentation
Checkpoint selection	Validation composite PL RMSE $+0.30$ DS RMSE $+0.25$ AS RMSE
Loss schedule	Large capacity probabilistic residual stage plus final residual fine tuning; see Table IX

TABLE IX
HARP-NET CKM LOSS SCHEDULE. THE PRE FINE TUNING PROBABILISTIC RESIDUAL STAGE IS PART OF THE FINAL PIPELINE, EVEN THOUGH THE REPORTED CHECKPOINT IS SELECTED AFTER THE DETERMINISTIC FINE TUNING STAGE.

Setting	Probabilistic residual stage	Final fine tuning
NLL / KL / moment / outlier	1.00 / 0.25 / 0.10 / 0.02	0 / 0 / 0 / 0
RMSE / MAE	1.50 / 0.10	1.00 / 0.05
Anchor / guard	0.05 / 0.10	0.01 / 0.02
Task weights	PL 1.0, DS 1.0, AS 1.0	PL 1.0, DS 0.25, AS 0.25
Learning rate	2×10^{-4}	5×10^{-5}

this is consistent with reducing rare high delay errors while introducing small local shifts.

The original numerical targets were 5 dB, 50 ns, and 20°. The final model clears all three on unseen cities. More importantly, it does so jointly: the same network predicts channel attenuation, DS, and AS instead of using separate specialist models.

a) Channel attenuation: The frozen path loss prior is already strong, reaching 1.9383 dB RMSE and 1.2319 dB MAE on the final test split. HARP-NET CKM reduces these to 1.6519 dB and 0.9298 dB. This is the most uniform gain: the residual model improves both squared error and typical absolute error, and it improves both LoS and NLoS pixels in Table XI. The closest external references are dense path loss radio map papers, but they still differ in split, height protocol, frequency, and target contract.

b) Angular spread: The angular spread prior reaches 13.7416° RMSE and 4.9890° MAE; the final model reaches 11.3854° RMSE and 5.0088° MAE. The near-constant MAE means the residual model mainly suppresses larger angular spread errors rather than shifting every pixel uniformly. The LoS gain is much larger than the NLoS gain, which is expected because the NLoS angular spread prior is already close to the target distribution. No reviewed external method reports the same dense CKM city holdout angular spread map task. The nearest papers predict link level or sparse angular spread channel parameters [23], [24], so the primary comparison is against the frozen prior and the thesis target.

c) Delay spread: The delay spread prior reaches 28.1023 ns RMSE and 7.3033 ns MAE; the final model reaches 26.5570 ns RMSE and 7.9897 ns MAE. This is the least uniform target: RMSE improves because rare high delay errors are reduced, while MAE slightly worsens because small local shifts appear in many easy pixels. The LoS/NLoS split confirms that the model improves both visibility cases in RMSE. Similar A2G delay spread predictors exist [22], [23], but they are not dense 513×513 CKM generators under city holdout, so they are used only as context.

TABLE X
FINAL MODEL TEST RESULTS ON UNSEEN CITIES. NEGATIVE Δ MEANS THAT THE RESIDUAL MODEL IMPROVES THE FROZEN PRIOR.

Target	Model RMSE	Prior RMSE	ΔRMSE	Model MAE	Prior MAE	ΔMAE	Unit
Attenuation	1.6519	1.9383	-0.2865	0.9298	1.2319	-0.3020	dB
Delay spread	26.5570	28.1023	-1.5453	7.9897	7.3033	+0.6864	ns
Angular spread	11.3854	13.7416	-2.3562	5.0088	4.9890	+0.0198	deg

TABLE XI
FINAL TEST RMSE SEPARATED BY LOS AND NLOS PIXELS.

Target	Scope	Model	Prior	Δ
Atten.	LoS	1.4675	1.7519	-0.2844
Atten.	NLoS	3.1482	3.5143	-0.3661
DS	LoS	25.4419	27.4851	-2.0432
DS	NLoS	29.2045	29.6157	-0.4112
AS	LoS	12.3510	15.2818	-2.9308
AS	NLoS	8.4424	8.6614	-0.2191

TABLE XII
VALIDATION VERSUS HELD OUT TEST CHECK FOR THE SAME SELECTED MODEL.

Split	Samples	Atten. RMSE	DS RMSE	AS RMSE
Validation	2750	1.6157	22.1061	11.3782
Test	2590	1.6519	26.5570	11.3854

B. LoS/NLoS and Split Diagnostics

Table XI separates valid pixels by the geometric visibility mask. The residual model improves the prior in both LoS and NLoS pixels for all targets. This is important because global RMSE could otherwise hide a failure in the physically hard NLoS subset.

The validation/test sanity check is also useful. The validation row evaluates the selected checkpoint on validation cities, while the test row is the real generalization result. Delay spread worsens more on test because the held out city mix contains harder high delay cases. The frozen delay spread prior shows the same validation to test worsening, confirming that this is a city mix effect rather than an evaluator change.

C. Prior and Split Breakdowns

The frozen priors are strong enough to deserve their own reporting. They are not simply baselines for the final neural model; they encode most of the stable geometry and therefore determine what residual problem the network sees. Because these prior calibrations are deterministic, non DL estimators, their standalone audits are reported separately from neural checkpoint selection. The path loss table uses the held out validation+test split after fitting the 10840 training references and evaluating 5340 held out references (4994 topology assigned rows). The spread table uses the final 14-city test split because the calibrated spread prior code and the final residual model use the same 10840/2750/2590 train/validation/test contract. Table XIII shows the height dependence of the calibrated path loss prior. This is not post hoc stratification: transmitter height is part of the prior estimator through the centred geometry, elevation angle, and height binned calibration, and it later conditions the residual neural model through FiLM. The trend is strongest in NLoS: the low altitude bin remains the hardest because many receivers are blocked by local urban structure, while high UAVs see fewer deep shadow pockets. Table XIV shows the matching spread prior pattern. Delay spread in particular collapses with altitude, from 39.90 ns overall in the lowest bin to 3.70 ns above 300 m. Combined with the empirical HDF5 histogram in Fig. 1, this means the high altitude rows are not better because they have more training support; the training split contains only 866 samples above 300 m. They are better despite that scarcity, which indicates a physically easier regime with fewer blocked links and weaker spread tails.

TABLE XIII
CALIBRATED PATH LOSS PRIOR RMSE BY TRANSMITTER HEIGHT ON THE HELD OUT VALIDATION+TEST SPLIT.

Height bin	Overall	LoS	NLoS	Samples
12–50 m	2.180	1.797	3.756	1211
50–150 m	1.936	1.785	3.228	2634
150–300 m	1.680	1.666	2.259	786
300–500 m	1.627	1.619	2.255	362

TABLE XIV
CALIBRATED SPREAD PRIOR RMSE BY TRANSMITTER HEIGHT ON THE FINAL 14-CITY TEST SPLIT.

Height bin	Metric	Overall	LoS	NLoS
12–50 m	DS	39.90	43.04	35.13
50–150 m	DS	27.17	26.72	28.29
150–300 m	DS	8.70	8.79	8.13
300–500 m	DS	3.70	3.80	2.76
12–50 m	AS	15.69	18.81	9.93
50–150 m	AS	14.19	15.89	8.41
150–300 m	AS	10.39	11.07	4.49
300–500 m	AS	8.85	9.34	2.85

TABLE XV
RMSE IMPROVEMENT OVER FROZEN PRIORS BY SPLIT. NEGATIVE VALUES MEAN THE RESIDUAL MODEL IMPROVES THE PRIOR.

Split	Samples	Atten.	DS	AS
Train	10840	-0.2895	-2.0157	-2.3578
Validation	2750	-0.2758	-1.3433	-2.4396
Test	2590	-0.2865	-1.5453	-2.3562
All	16180	-0.2867	-1.8406	-2.3710

The path loss prior also varies by topology. Its overall RMSE ranges from 1.43 dB in open sparse low rise maps to 2.97 dB in dense block high rise maps. The NLoS branch is hardest in sparse vertical and dense high rise regimes, reaching 3.91 dB and 4.09 dB; the overall pixel weighted prior remains 1.92 dB.

The final residual model improves the priors consistently across data splits, not only on the validation split used for model selection. Table XV reports the model minus prior RMSE deltas. The test row is the main generalization claim. The all city row is not used for model selection, but it checks that the gain is not a held out split artifact.

The all city morphology/height audit groups samples by broad urban form (`open`, `mixed`, `dense`) and antenna height

TABLE XVI
ALL CITY FINAL MODEL RMSE DELTAS BY MORPHOLOGY/HEIGHT AUDIT GROUP. NEGATIVE VALUES MEAN THE FINAL MODEL IMPROVES THE FROZEN PRIOR.

Group	Samples	Atten.	DS	AS
dense/high	1957	-0.3602	-0.6473	-2.6405
dense/low	2018	-0.3359	-1.5779	-2.3994
dense/mid	2055	-0.3288	-1.4074	-2.8587
mixed/high	2156	-0.2931	-0.6794	-2.3186
mixed/low	2060	-0.2745	-2.8379	-2.5845
mixed/mid	2094	-0.2781	-2.0522	-2.8200
open/high	1272	-0.2721	-0.5428	-1.2874
open/low	1269	-0.2363	-3.1170	-2.0271
open/mid	1299	-0.2941	-1.5789	-1.7804

bin (`low`, `mid`, `high`), using the same 58.12 m and 103.85 m thresholds as the frozen prior calibration keys. These rows are not nine separately trained final neural networks. They are evaluation buckets that mirror the regime structure used by the frozen priors and fallback tables: morphology is derived from building density/height thresholds, and the height tag comes from the transmitter altitude range. This is a stricter check than the global row because it asks whether the same residual model helps across the deployed prior regimes rather than only in one easy subset. Table XVI reports RMSE deltas for the nine groups. All entries are negative, so the residual correction improves every morphology and height group for all three targets.

D. Metric Interpretation

The reported model is deterministic at evaluation time: all tables compare the continuous HARP-NET CKM point maps with the stored ray-traced target maps over valid ground receiver pixels. The residual head is distribution aware during training because attenuation, delay spread and angular spread have different LoS/NLoS regimes and tail behaviour, but the final claim is the RMSE and MAE of the exported point prediction.

For spread targets, quantized labels make visual and RMSE interpretation harder. The released delay and angular maps are integer valued; smooth predictions can be penalized across staircase like target boundaries even when the morphology is plausible. This does not invalidate RMSE, but it warns against overinterpreting small local residuals.

E. External Metric Context

Table XVII places the final channel attenuation result beside the closest available path loss side references. The table is deliberately compact and cautious: datasets, height assumptions, target definitions, and splits differ. No scalar indoor or link level spread predictor is used as a direct dense map leaderboard.

Table XVIII gives the corresponding spread side context. These rows are not direct rankings, because the external works report scalar or link level channel parameters rather than dense CKM maps. They are included so that the DS/AS results are still positioned against the nearest published spread prediction literature.

The fair conclusion is narrow but useful. The final channel attenuation number is in the same band as strong dense radio map references. It is numerically below the recent Gao et al. outdoor segmentation results, but that comparison remains cross dataset scale context rather than a leaderboard result. The spread results now have explicit nearby literature context in Table XVIII, but should still be interpreted primarily against same protocol internal baselines because no external dense CKM channel attenuation/DS/AS benchmark was found.

TABLE XVII
COMPACT COMPARISON WITH NEARBY PATH LOSS SIDE METRICS.

Reference	Attenuation / PL side metric	Reading / caveat
RadioUNet [5]	1.6 dB–3.1 dB equivalent PL RMSE	Final attenuation result is in this band; no transmitter height or spread targets.
RadioGUNet [10]	1.304 dB DPM; 1.936 dB IRT	Final attenuation result is between those values; outdoor RadioMapSeer setting.
PMNet / ICASSP [7], [8]	approx. 1.38 dB converted score	Final attenuation result is close but higher under a different benchmark contract; scale context only, no joint spread prediction.
Gao et al. [12]	5.59 dB ICASSP 2023; 12.19 dB ITU 2024	Final attenuation result is numerically lower, but this is scale context rather than a like for like win: recent corridor weighted outdoor PL predictor, no variable height UAV or joint spread outputs.
ReVeal [15]	1.95 dB outdoor sparse measurement RMSE	Final attenuation result is in the same range; sparse mapping rather than dense generation.
RMTransformer / PathFinder [9], [11]	$\approx$ 1.82 dB / $\approx$ 1.19 dB converted references	Useful scale checks, but random/specialized splits and different assumptions.
AIRMap [13]	Path gain error $<$ 4 dB against ray tracing	Same broad digital twin motivation, but path gain and a fixed 200×200 input grid with variable physical extent.
Final model (ours)	**1.6519 dB attenuation; 26.5570 ns DS; $11.3854°$ AS**	**Strict CKM city holdout, centred variable height UAV, dense 513×513 channel attenuation/DS/AS.**

F. Qualitative Residual Behavior

Fig. 6 is the qualitative check. The model receives the frozen prior and predicts a correction. The useful visual evidence is not merely that the prediction resembles the target; it is that the correction row tends to correlate negatively with the prior error row, which is the expected behavior of a HARP-NET CKM residual corrector.

G. Runtime

The final generator was compared against the supplied MATLAB ray tracing path on Barcelona layouts at full 513×513 resolution on the same laptop: an AMD Ryzen 5 5600H CPU and an NVIDIA RTX 3050 Ti Laptop GPU. The surrogate includes LoS/NLoS mask generation, frozen priors, final neural prediction, and default numeric export after the checkpoint is loaded once. It runs in 0.88 s per sample versus 99.89 s for the local ray tracing path, a $113.7\times$ speedup under this measurement setup. The ray tracing row uses the CPU path; the surrogate row uses the laptop GPU with mixed precision.

H. Interpretation and Limitations

The final result is best interpreted as evidence for a hybrid CKM design rather than as a claim that one backbone solves

TABLE XVIII

SPREAD SIDE COMPARISON WITH NEARBY CHANNEL PARAMETER PREDICTION WORK. EXTERNAL ROWS GIVE TASK AND METRIC CONTEXT, NOT A DIRECT DENSE MAP LEADERBOARD.

Reference	Spread side reported output	Relation to this work	Why it is not direct
Yang et al. [22]	A2G mmWave delay spread prediction with RF/kNN; lower RMSE than a lognormal contrast model.	Closest early A2G delay spread ML precedent.	Scalar DS only; no AS, no dense map, no CKM city holdout.
Huang et al. [23]	Transformer UAV predictor; merged data RMSE 4.577 ns for RMS DS and 1.549°, 0.375°, 0.864°, 0.074° for RMS AAOA, EAOA, AAOD and EAOD.	Recent UAV mmWave model with delay and angular spread outputs.	Per link characteristics from coordinates/frequency/location tags, not dense 513×513 AS/DS maps from topology images.
Mi et al. [24]	Regression PointNet for indoor 60 GHz measurements; predicts RMS DS, azimuth AS, PL and K-factor; average AS RMSE 7.39°.	Measurement based DL reference including DS and AS.	Indoor point cloud task with link level samples, not outdoor UAV CKM generation.
Final model	**26.5570 ns DS RMSE and 11.3854° AS RMSE.**	**Dense channel attenuation/DS/AS maps under the final city holdout contract.**	**Should be judged mainly against the frozen spread priors and thesis targets under the same protocol.**

TABLE XIX

RUNTIME COMPARISON FOR ONE FULL RESOLUTION BARCELONA CKM SAMPLE ON THE SAME RYZEN 5 5600H / RTX 3050 TI LAPTOP HARDWARE.

Method	Timed path	Time
MATLAB RT	Pixel receivers outside buildings; attenuation/DS/AS parsed from paths; CPU path	99.89 s
Final generator	Masks, priors, attenuation/DS/AS prediction, numeric export; CUDA mixed precision	0.88 s

radio mapping in general. The strongest technical signal is that the residual model improves a high quality frozen prior across train, validation, test, and every morphology/height audit group. That makes the gain harder to dismiss as overfitting to a single easy city mix. At the same time, the modest size of the delay spread gain is a useful warning: once the calibrated prior has absorbed the large geometry trend, the remaining spread error is dominated by rare tails and target quantization, not by a smooth missing term. The residual errors that remain are most likely where low UAV elevation, dense local clutter, and NLoS support coincide, because those pixels depend on the particular blocked/scattered path selected by the ray tracer rather than only on broad radial or morphology statistics.

Several limitations remain. First, the labels come from ray tracing rather than field measurements, so the result is simulator agreement, not over the air validation. Second, target quantization limits how much sub unit error structure can be learned. This is not a limitation of the output layer: the model outputs continuous channel attenuation, DS, and AS values. It is a limitation of the stored labels. For channel attenuation, a 0.9298 dB final MAE against labels quantized at about 1 dB is already close to the label step, even though the RMSE still reveals harder tails. Third, the final model is selected under one city holdout split and one final training seed. Fourth, the dataset fixes frequency, antenna pattern, and simulator assumptions; deployment would require conditioning or calibration across those variables.

V. CODE AVAILABILITY

The reproducibility record is public as live source repositories. The citations below identify the public source locations and snapshot commits or tags used for this manuscript. The standalone CKM Generator repository contains the Streamlit interface, command line generator, frozen calibration files, final model runtime code, and the published checkpoint path used by the demonstrator [37]; it is the inference/demo entry point. The final training and evaluation implementation is collected separately in the final code repository [38]; it is the source record for reproducing the paper metrics. The reported tables are tied to the selected `best_model.pt` checkpoint and its exported comparison summary and deep breakdown JSON files, rather than to a later interactive demo run. For traceability, the LaTeX thesis source and project source archive are also available in public repositories [39], [40]. Large data and model artifacts are treated as external artifacts or Git LFS objects rather than ordinary source files.

VI. CONCLUSION

This paper presents a dense variable height UAV CKM surrogate that jointly predicts channel attenuation, delay spread, and angular spread on unseen cities. Under the strict city holdout contract, the final model reaches 1.6519 dB, 26.5570 ns, and 11.3854°, respectively, while using one centred UAV generator across the sampled height range. The result is not only a neural architecture result. Its main contribution is a hybrid recipe: first build strong train only calibrated priors for the structure that is predictable, then learn bounded residual distributions around those anchors.

The calibrated priors are themselves an important part of the method. They put together coherent two ray LoS path loss, height dependent calibration, morphology aware NLoS correction, visibility switching, and log domain DS/AS spread statistics. These frozen anchors already explain most radial, visibility, height, and spread scale structure; the final residual model then improves them by 0.2865 dB, 1.5453 ns, and 2.3562° RMSE on the held out test set. Continuous transmitter height is handled through both the height aware priors and a sinusoidal FiLM conditioning path, so the same model covers the varying height regime without per altitude retraining.

The distribution first diagnosis remains the key modelling lesson. NLoS channel attenuation residuals are mixture-like and prone to mode collapse, while DS/AS targets are quantized and spike plus tail rather than single smooth residuals. GMM style residual heads and histogram checks therefore matter because they prevent visually plausible maps from hiding collapsed marginals. The remaining spread errors are consistent with this interpretation: the large scale DS/AS structure is usually recovered, but rare high spread pixel groups can still be missed, and some false positive high spread patches remain.

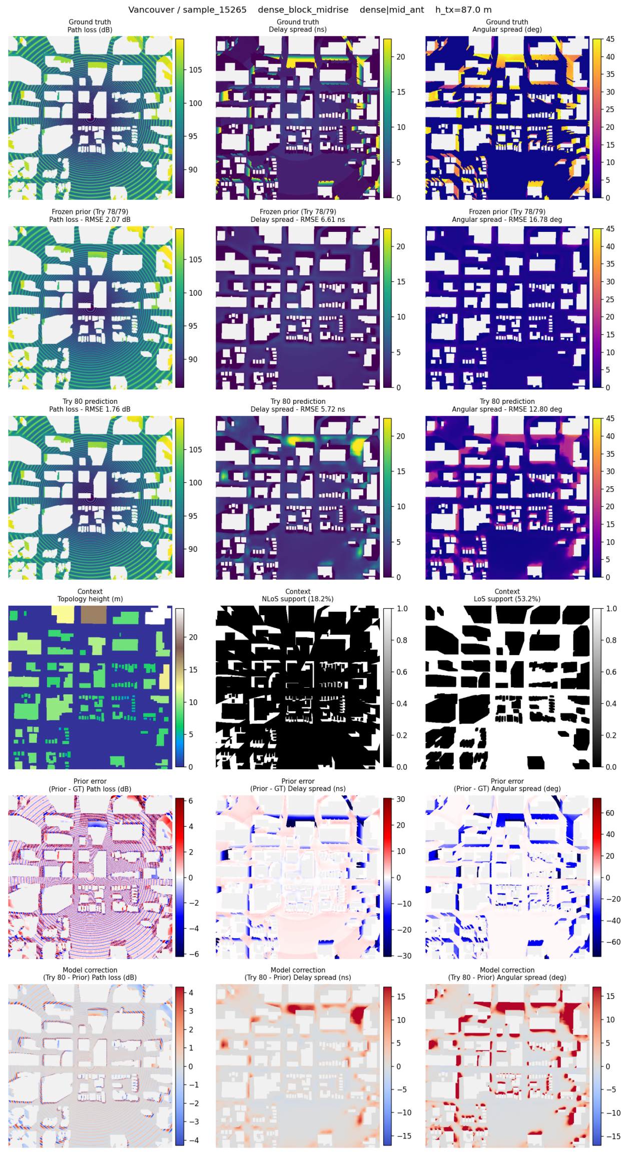

Fig. 6. Representative final model inspection panel for a dense, mid antenna held out test sample in Vancouver. Columns are channel attenuation, delay spread, and angular spread. Rows show ground truth, frozen prior, final prediction, context/support, prior error, and model correction.

The main practical conclusion is narrow and useful. For this fixed ray traced CKM simulation contract, HARP-NET CKM is accurate, fast, and interpretable enough to reduce repeated ray traced map generation for many planning experiments, achieving a measured 113.7× speedup over the local MATLAB ray tracing path. The result is a reproducible surrogate for a defined dataset, split, carrier, antenna setup, and receiver mask, and its clearest lesson is methodological: diagnose the target distribution, freeze the stable structure in auditable priors, and spend neural capacity on the residual structure that the priors cannot explain.

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